

Meridian Supply Chain Intelligence RAG System

HCLTech Assignment 2 — Comprehensive Final Technical & Audit Report

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GitHub Repository:	goutham-11-16/Supply-Chain-RAG	Demo Video URL:	Google Drive (Link below)
Vector Store:	ChromaDB (supplychain_rag)	Indexed Vectors:	22 Chunks (Size: 1200, Overlap: 150)

1. Project Overview

This project establishes a production-quality, document-grounded Retrieval-Augmented Generation (RAG) system for **Meridian Components Pvt. Ltd.**, an automotive electronics manufacturer operating assembly plants in Chakan (Pune) and Hosur (Tamil Nadu). Meridian's procurement team handles complex buyer queries regarding supplier performance metrics (on-time delivery, defect PPM, line stoppages) and corporate policy rules (PO approval tiers, quality cost recovery under Clause 6.3, and safety stock formulas).

2. System Architecture

PDF Documents in data/  PyPDF Loader  Recursive Splitter (1200/150)  text-embedding-3-small  GPT-4o LLM (0.1)  Top-K Search (1-12)  ChromaDB (supplychain_rag)

3. Document & Dataset Details

Document Name	Document Type	Pages	Key Content Covered
Meridian_Procurement_Policy_Handbook_v4.2.pdf	Corporate Policy	3 Pages	Approval authority tiers, supplier classification, Clause 6.1 OTD warnings, Clause 6.3 quality penalties ( 120/unit), Section 8 safety stock formulas.
Meridian_Supply_Chain_Review_Q1_FY2025-26.pdf	Operational Review	3 Pages	Q1 scorecard metrics (OTD, PPM) for 5 key suppliers, 7 line stoppage events (41 hrs total), single-source microcontroller risks, freight lanes.

4. Retrieval Strategy

Top-K similarity search queries ChromaDB with `text-embedding-3-small` (default K=6, slider range 1–12). Smart candidate merging and full-content string deduplication ensure context balance across both PDFs.

5. Cross-Document Reasoning

- **Kaveri Metals:** Combines Review Q1 (88.1% OTD, 1,150 PPM) with Handbook Clause 6.1 warning & Clause 6.3 120/unit rework debit + 100% inspection floor.
- **Trident Circuit Boards:** Combines Review Q1 (640 PPM defect rate) with Handbook Clause 6.3 penalties.
- **Shenzhen Rui Electronics:** Combines Review Q1 (Highest spend 21.9 cr, 76.0% OTD) with Handbook Clause 7.1 (mandatory dual-sourcing) and Vietnam second-source actions.

6. Policy & Penalty Calculator

Deterministically evaluates Clause 6.1 (OTD < 90% warnings), Clause 6.2 (Rating Bands A–D), Clause 6.3 (■120/unit rework recovery debit & 100% inspection floor), and Section 8 Safety Stock math: $SS = \max(Lead\ Time \times 0.25, Floor\ Days)$.

7. Comprehensive 20-Test Suite Benchmark

All 20 evaluation test cases passed with 100% precision. Evaluates single-doc queries, cross-doc reasoning, policy math, and trap refusal.

8. Official 3-Minute Demonstration Video

Video URL: <https://drive.google.com/drive/folders/1ggQuVtzNUMUtFqDbNb10j7WM9xxqFDpd?usp=sharing>

Walkthrough Breakdown:

- 0:00 - 0:20: Introduction to Meridian PDFs.
- 0:20 - 1:00: Database ingestion, reporting 22 unique vector chunks in ChromaDB.
- 1:00 - 2:30: Live cross-document Q&A; (Kaveri Metals Q5 & Safety Stock Q7) with dual-column page citations.
- 2:30 - 3:00: Trap refusal demonstration & developer debug tool inspection.

9. Hallucination / Trap Test

Query: 'What is the annual salary of the Head of Procurement?'
Output: **'The information is not available in the uploaded documents.'** (0 chunks referenced).

10. Development History & Fixes

- Corrected deduplication key to compare full content ``doc.page_content.strip()``, preserving 1-to-1 Top-K counts.
- Resolved Windows file locking via Chroma's native ``existing.delete_collection()`` reset call.

11. Final Verification Checklist

Verification Item	Requirement	Status
Both Meridian PDFs Indexed	22 unique vector chunks in 1 ChromaDB collection	■ VERIFIED
Top-K Range Control	Exact 1-to-1 retrieval for K = 1–12 (default 6)	■ VERIFIED
Cross-Document Reasoning	Synthesizes Review Q1 metrics + Handbook policy rules	■ VERIFIED
Source Citations	Cites document title and page number for every fact	■ VERIFIED
Demo Video Integration	Live Google Drive URL embedded in README & Report	■ VERIFIED
API Key Security	.env listed in .gitignore; zero exposed secrets on GitHub	■ VERIFIED

12. Conclusion

The Meridian Supply Chain Intelligence RAG System is fully built, tested, documented, and verified by **Esambadi Goutham Reddy**.

Report generated autonomously for HCLTech Assignment 2 Submission Verification.